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SUPPLEMENTARY MATERIALS

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BIRD FLIGHT

Frigate birds track atmospheric conditions over months-long transoceanic flights

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Understanding how animals respond to atmospheric conditions across space is critical for understanding the evolution of flight strategies and long-distance migrations. We studied the three-dimensional movements and energetics of great frigate birds (*Fregata minor*) and showed that they can stay aloft for months during transoceanic flights. To do this, birds track the edge of the doldrums to take advantage of favorable winds and strong convection. Locally, they use a roller-coaster flight, relying on thermals and wind to soar within a 50- to 600-meter altitude band under cumulus clouds and then glide over kilometers at low energy costs. To deal with the local scarcity of clouds and gain longer gliding distances, birds regularly soar inside cumulus clouds to use their strong updraft, and they can reach altitudes of 4000 meters, where freezing conditions occur.

The movement of animals is driven by processes that act across multiple spatial and temporal scales. Long-distance movements such as the migrations of birds have evolved in response to large-scale environmental gradients (*1*). In particular, atmospheric conditions play a large role in determining the efficiency of migratory routes, whose consistency over years has allowed evolutionary processes to act at population levels (*2*). At smaller time and spatial scales, long-range movements have to constantly be adjusted to local conditions, in particular to minimize energy expenditure (*3*, *4*). These long movements or migrations can be done over inhospitable areas as different as deserts, high mountains, or oceans, which come with specific environmental constraints to which birds need to behaviorally and physiologically adapt their flight strategies (*5*, *6*). How these long restless flights can be energetically achieved has attracted much interest, but remains largely unknown because of the inherent difficulties of studying such behaviors in situ.

Biologists have long been attracted to locomotor extremes because they provide clear examples from which information about structure-function relationships can be drawn (*7*). Among birds, frigate birds are extreme in many aspects of their life history, including having the lowest wing loading, with a specialized capacity for soaring flight (*8*). They are also unusual seabirds because their feathers are not waterproof and their legs are small, so they are unable to land on the sea surface even though they feed exclusively at sea. They deal with these conflicting constraints by staying aloft for days when they are foraging from their nest when breeding (*9*). Probably as a consequence of these extreme attributes, frigate birds have the longest period of parental care in birds, suggesting a long period of learning to acquire flight and foraging abilities in early life (*10*). Their ability to remain airborne continuously for days is probably possible because of the capability of frigate birds to use thermals over the sea as a main energy source for soaring (*11*, *12*).

We asked how frigate birds can perform long migrations over oceans without landing and whether oceanic thermals are reliable enough in space and time to allow birds to stay airborne over long periods. To address these questions, we investigated the movement of frigate birds at several spatial scales with regard to (i) how frigate birds make use of large-scale weather systems to perform long-range movements, and (ii) how flight dynamics and energetics at a finer scale contribute to these long ranges.

We studied the three-dimensional movements and energetics of frigate birds on Europa Island (Fig. 1) between 2011 and 2015 (*13*). To study large-scale migratory movements, 24 adults and 25 juvenile birds were equipped with solar-powered Argos transmitters (*13*). To study the relationship between heart rate, activity (flapping frequency), and behavior (ascent rates and horizontal speed), 11 adult females were equipped with external custom-designed loggers measuring triaxial acceleration and electrocardiography and a Global Positioning System (GPS) device (*13*). To study movements, activity, and ambient temperature, 37 adult females and males were equipped with solar-powered GPS accelerometers, whose data were recovered regularly by an automatic recording station (*13*).

During the southwest Indian monsoon from June to October, strong trade winds occur in the southern Indian Ocean and cross the equator to form southwest winds in the northern Indian Ocean (*14*) (Fig. 1). During this season, adult frigate birds finishing the breeding season left Europa and migrated northward to take advantage of the southerly winds. They settled on roosting sites in

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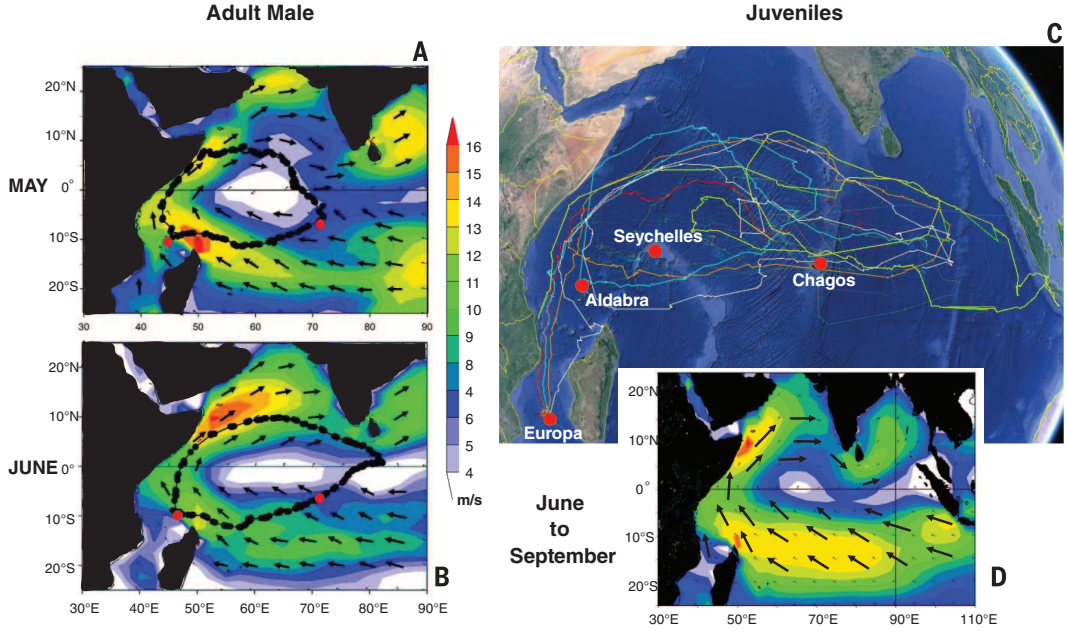


Fig. 1. Movements of adult and juvenile frigate birds in relation to wind conditions in the Indian Ocean. (Left) Two successive clockwise movements from Aldabra Island (Seychelles) of an adult male great frigate bird (no. 138502) in relation to wind strength (in meters per second, color scale) and direction (arrows). (**A**) In May 2015, a 24-day foraging trip around the doldrums (shown by the absence of wind, in white), with 1 day of rest in Chagos. (**B**) In June 2015, a 28-day foraging trip, with a 36-hour rest in Chagos. (**C**) Movements between June and September 2015 of six young frigate birds fledged from Europa Island, moving around the doldrums zone. (**D**) Climatology of wind speed and direction (average values over 4 months) in June to September 2015, showing the average position of the doldrums (white) on the equator.

the Seychelles from where they foraged for months. Some adults performed long looping movements around the equator, where a belt of converging air and wind occurs, with no wind in the center, named the doldrums zone by ancient mariners (Fig. 1). On successive loops, adults closely followed the edges of the doldrums, which oscillate longitudinally (Fig. 1, A and B). Birds stayed continuously on the wing for periods lasting up to 48 days and traveled on average 420 ± 220 km daily.

Young frigate birds left their birthplace at the same time as adults, but independently of their parents. They crossed the equator and turned eastward to enter into a circular transoceanic movement into the wind belt around the doldrums (Fig. 1, C and D). During these dispersive movements, juvenile birds stayed continuously aloft for flights lasting up to 2.1 months (average maximum time spent aloft, 41.2 ± 15.1 days, n = 8 birds). They travelled on average 450 ± 220 km daily. They episodically stopped on isolated islands such as Chagos, islets off Indonesia, or on islets of the Seychelles archipelago for very short rests (8 to 48 hours) before continuing their large-scale wandering movement tracking the edge of the doldrums (Fig. 1C and fig. S1). They flew at altitudes ranging between the sea surface and 3000 m, but mainly between 0 and 600 m (fig. S2).

To understand how frigate birds are able to stay aloft for such long periods, we studied their flight dynamics and energetics during 2- to 15-day foraging trips from Europa. Breeding frigate birds travelled on average 410 ± 142 km per day (n = 18 birds), mainly during the daytime, traveling over shorter distances at night (Fig. 2). Two clear behavioral modes were identified during movements at sea. Traveling occurred with high ground speeds and low wing beat frequencies (82 ± 9% of travelling time with no or rare wing beats), with birds remaining at altitudes ranging from 30 to 2000 m, reaching up to 4120 m. Foraging can only occur when birds descend close to the sea surface (altitudes 0 to 30 m), and during these periods they are very active, flapping during 75 ± 18% of the foraging phase (Fig. 2). Active foraging occurred only episodically (10 ± 7% of time at sea), indicating rare feeding opportunities, mainly during the day (86.4% of bouts of active foraging occurred during the daytime).

When in flight at sea, heart rate was on average 203 ± 84 beats min^{-1} but varied extensively (Fig. 2 and figs. S3 and S4), occasionally attaining values as low as when resting on the nest (71 ± 25 beats min^{-1}, range 57 to 215). Heart rate and dynamic body acceleration were generally well correlated (*13*) (figs. S3 and S4); therefore, we used dynamic body acceleration, measured on all individuals, as the main proxy for energy expenditure. Whereas active foraging is very costly for frigate birds, requiring high dynamic body acceleration and heart rates, traveling periods have a remarkably low energy expenditure, with few wingbeats (Fig. 2 and fig. S3), suggesting that overall field metabolic rate during months at sea is likely to be exceptionally low (*13*) (figs. S5 and S6). Excluding periods of active foraging close to the surface, dynamic body acceleration was the lowest at altitudes between 300 and 600 m (fig. S7), indicating an optimal altitude for traveling at low cost.

Traveling at low cost is achieved by successive climbs, mainly through soaring with no or few wing flaps and low heart rate, and descents, by gliding (Figs. 2 and 3). A close examination of flight paths shows that when soaring, birds move with the wind (fig. S8), using circling movements to soar (Fig. 3C) in thermals below cumulus clouds where rising air creates updrafts (*15*). Because of the strong trade winds, they drift with the wind while climbing (Fig. 3), resulting in "wind-drift circling soaring." Conversely, when gliding, they preferentially fly with side winds and achieve the highest ground speeds (Figs. 2 and 3 and fig. S8). The resulting movement is a complex zig-zagging, roller-coaster movement, with an average altitude xgain of 59.1 ± 43.8 m per kilometer covered (ground distance); i.e., 15.4 ± 3.0 km climbed daily. These vertical movements take place generally up to 600 to 700 m, corresponding to the base of the cumulus clouds that is relatively constant throughout the trade wind zone (*16*).

However, birds regularly climbed up to 1000 to 2000 m, with a maximum of 4120 m (fig. S9). The frequency of climbs at altitudes higher than 700 m increased throughout the day to peak during the first hours of night (fig. S10). Climbing to high altitudes can be separated into a phase of slow climb up to the base of the cumulus clouds at 600 to 700 m, followed by a more rapid climb to 1600 m (Fig. 2D) or higher. This second phase of the ascent is performed without flapping the wings (Fig. 2D); i.e., in pure soaring flight and with climb rates reaching 4 to 5 m s^{-1} at 2000 to 3000 m (fig. S11). Ascent to high altitude can only take place inside cumulus clouds, where updrafts reach 5 m s^{-1} and

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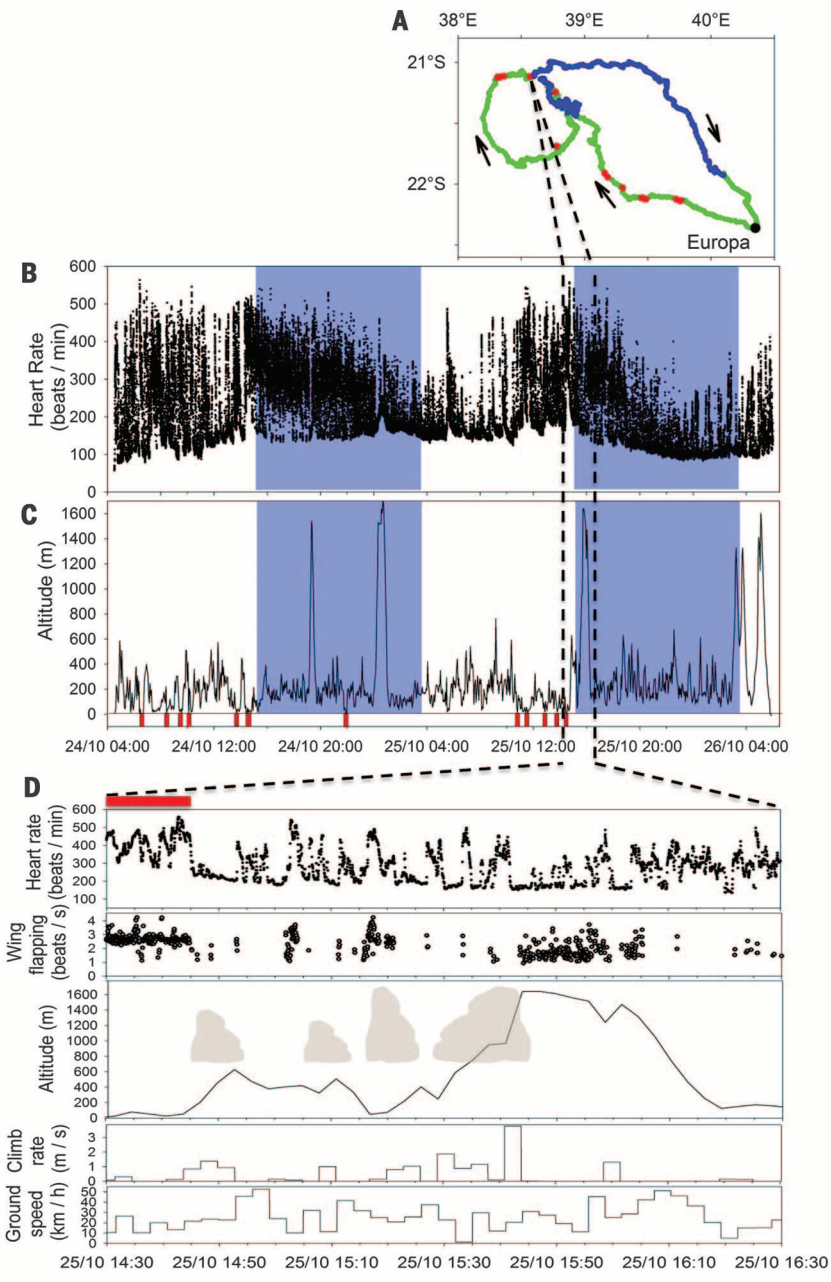


Fig. 2. Movement, changes in altitude, heart rate, and flight parameters during a 2-day trip at sea. (**A**) 1130-km-long movement from Europa during the daytime (green) and night (blue) with foraging bouts in red, with recorded (**B**) heart rate and (**C**) altitude. Shaded blue areas represent nighttime, and red blocks represent foraging bouts. (**D**) A 2-hour period during the trip at sea, with an active foraging phase followed by a traveling phase, showing the changes in altitude and corresponding heart rate, wing-flapping rate, climb rate, and ground speed. On the altitude panel, the predicted presence of cumulus clouds is indicated (gray).

are strong enough to provide such climb rates (*15*). During the gliding phase made outside the clouds, the minimum sink rate was 23.6 ± 19.1 m of ground distance covered per meter lost between 500- to 700-m altitudes, compared to 14.3 ± 11.7 m at higher altitudes ($F_{1,7}$ = 8.4, P = 0.045).

Our study shows that frigate birds can remain almost indefinitely on the wing by tracking, at a basin-wide scale, the wind belt around the doldrums, an atmospheric feature whose location is predictable. Locally, they display a specific flight strategy based on an energy-efficient use of convection and wind. By using wind-drift circling soaring and long periods of gliding, frigate birds are able to simultaneously use convection and wind as energy sources and move over extensive distances at low energy costs. They favor altitudes between 50 and 600 m, where atmospheric conditions are optimal for low-cost flight; i.e., steady winds and updrafts from convection under clouds (Fig. 4). These altitudes are also convenient to spot feeding opportunities from long distances away during the daytime; birds then descend close to the surface to forage actively when feeding opportunities have been detected (*9*).

Although birds are not thought to carry out intentional, sustained cloud climbs (*17*, *18*), our study shows the ability of frigate birds to frequently ascend to very high altitudes inside clouds. At an altitude of 4000 m, air temperatures are negative and air density and oxygen availability are almost half of those at sea level (*17*), suggesting that this tropical bird encounters extreme conditions at such altitudes. Cumulus clouds and cloud fields are considered to be randomly distributed in space in the trade wind zone (*16*). In these conditions, climbing higher than 1000 m presents a fundamental advantage by allowing frigate birds to cover much longer distances by gliding to reach the next updraft under clouds; for example, >60 km of glide from 4000 m as compared to 17 km when starting a glide just at the base of cumulus clouds. Therefore, when clouds are sparsely distributed, birds can adjust their gliding distance by climbing higher to avoid the risk of switching to costly flapping flight. Juvenile individuals are able to master the flight strategy of adults as soon as they become independent. When they leave their birthplace, they all head north to reach the equator and circle the entire Indian Ocean. This stereotyped movement suggests a genetically encoded behavior that brings young individuals directly to a predictable, favorable, and large-scale atmospheric feature located thousands of kilometers from their birthplace.

Great frigate birds are the only birds other than swifts (*19*) to be able to stay aloft for months. Long periods in continuous flight are interrupted by very short periods of rest on land, suggesting that frigate birds might sleep while airborne (*20*). Periods of low activity (no flapping) occur mainly during soaring episodes and may allow sleep. However, periods of completely motionless (no flapping at all) flight, potentially corresponding to periods of sleep, are relatively short, (~2 min, never exceeding 12 min). Animals such as frigate birds may have evolved the ability to dispense with sleep when ecological demands favor wakefulness such as during extended flights (*21*), but studies are needed to determine how they sleep during much longer-lasting flights

Frigate birds clearly encounter several atmospheric challenges during their movements at sea, such as low temperatures, low air density and oxygen levels during high climbs, and the unpredictable distribution of cumulus clouds at small scale, together with the presence of powerful cyclones in their optimal range. This dependence on atmospheric systems could make them particularly sensitive to future climate changes, along with some other seabirds (*22*). Climate models for the tropical ocean forecast an increase in the intensity of tropical storms and of convections around the equator, where the doldrums and strong convections occur (*23*). More variable atmospheric

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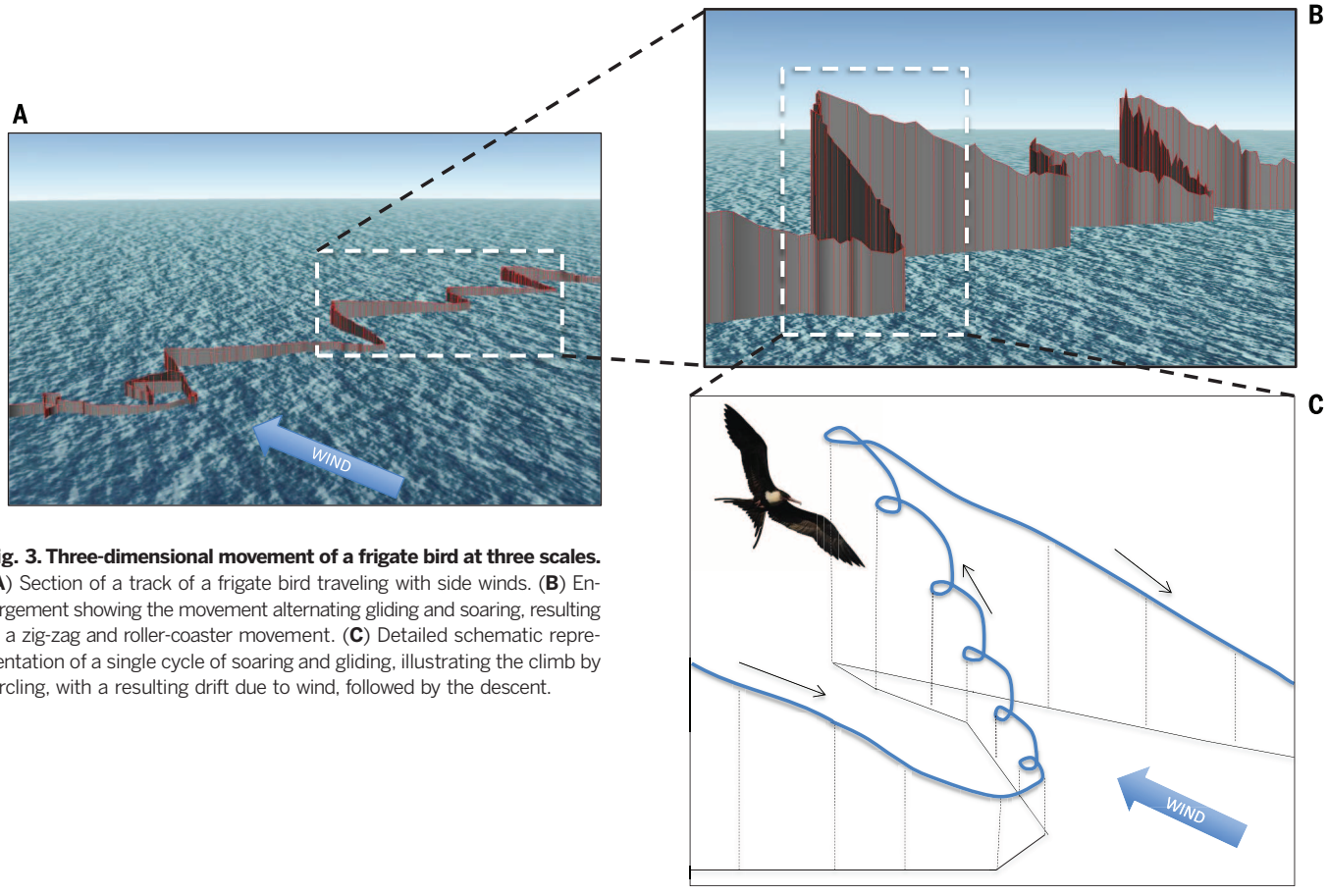


Fig. 3. Three-dimensional movement of a frigate bird at three scales. (**A**) Section of a track of a frigate bird traveling with side winds. (**B**) Enlargement showing the movement alternating gliding and soaring, resulting in a zig-zag and roller-coaster movement. (**C**) Detailed schematic representation of a single cycle of soaring and gliding, illustrating the climb by circling, with a resulting drift due to wind, followed by the descent.

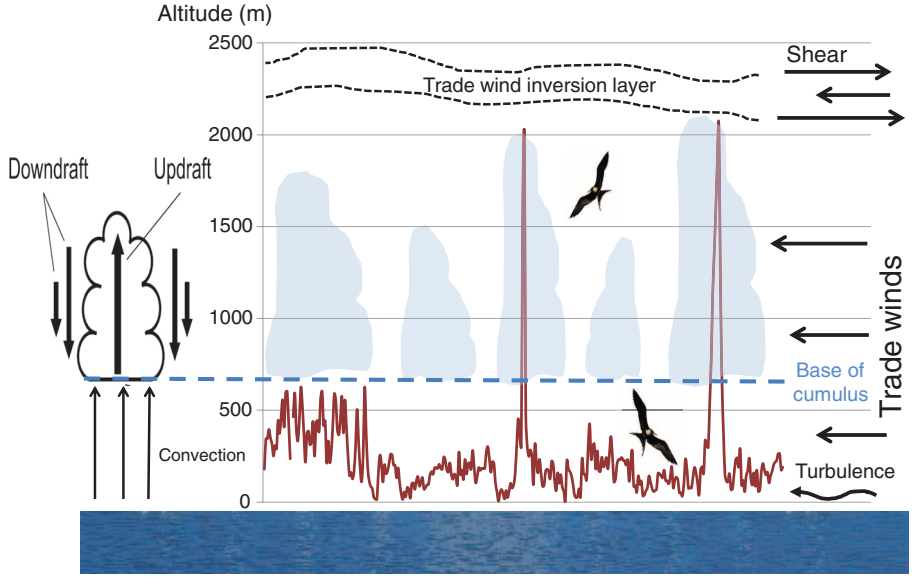


Fig. 4. Altitudinal movement of a frigate bird in relation to potential atmospheric conditions. The traveling flight is performed between 30 and 600 to 700 m altitude in a band with regular winds, avoiding the turbulence close to the surface. The bird occasionally climbs to 2000 m within cumulus clouds that form by convection, whose base is at 600 to 700 m and whose vertical extension is limited by the inversion layer where strong shear occurs. A schematic presentation of the updrafts and downdrafts characteristic of cumulus clouds is shown at left (*24*).

conditions in the future may become too challenging for a species that already seems to encounter extreme conditions during its lifetime movements.

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SUPPLEMENTARY MATERIALS

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TRANSCRIPTION

Visualization and analysis of gene expression in tissue sections by spatial transcriptomics

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Analysis of the pattern of proteins or messenger RNAs (mRNAs) in histological tissue sections is a cornerstone in biomedical research and diagnostics. This typically involves the visualization of a few proteins or expressed genes at a time. We have devised a strategy, which we call "spatial transcriptomics," that allows visualization and quantitative analysis of the transcriptome with spatial resolution in individual tissue sections. By positioning histological sections on arrayed reverse transcription primers with unique positional barcodes, we demonstrate high-quality RNA-sequencing data with maintained two-dimensional positional information from the mouse brain and human breast cancer. Spatial transcriptomics provides quantitative gene expression data and visualization of the distribution of mRNAs within tissue sections and enables novel types of bioinformatics analyses, valuable in research and diagnostics.

Tissue transcriptomes are typically studied by RNA-sequencing (RNA-seq) (*1*) of homogenized biopsies, which results in an averaged transcriptome and loss of spatial information. The positional context of gene expression is of key importance to understanding tissue functionality and pathological changes. Several strategies have recently been developed with this aim (*2*–*5*), but they have limitations in the number of transcripts that can be analyzed, rely on rich preexisting data sets, and/or are costly and labor-intensive, and none of them are operational in the standard research and diagnostic setting of regular histological tissue sections.

We asked whether it would be possible to introduce positional molecular barcodes in the complementary DNA (cDNA) synthesis reaction within the context of an intact tissue section before RNA-seq. We first assessed whether it was feasible to generate cDNA from messenger RNA (mRNA) in tissue sections on a surface. We immobilized reverse-transcription oligo(dT) primers on glass slides and placed on the slides sections of adult mouse olfactory bulb, a brain region with clear histological landmarks and ample gene-expression reference data. The tissue was fixed, stained, and imaged (Fig. 1A) (*6*).

After permeabilization, we added reverse-transcription reagents on top of the tissue. We used fluorescently labeled nucleotides to visualize the synthesized cDNA (Fig. 1A and fig. S1). The tissue was then enzymatically removed, which left cDNA coupled to the arrayed oligonucleotides on the slide (*6*). The fluorescent cDNA showed a pattern in detail corresponding to the tissue structure revealed by the general histology (Fig. 1, B and C), and the cDNA was strictly localized directly under individual cells (Fig. 1, D to G′). By comparing the hematoxylin-and-eosin and fluorescent signals, we could measure the average distance of diffusion outside the border of a cell to 1.7 ± 2 μm (mean ± SD) (fig. S1, E to H).

The realization that it is possible to capture mRNA in tissue sections with minimal diffusion and maintained positional representation motivated us to array oligonucleotides with positional barcodes (Fig. 2A), and we denoted this strategy "spatial transcriptomics." We deposited ~200 million oligonucleotides in each of 1007 features, with a diameter of 100 μm and a center-to-center distance of 200 μm, over an area of 6.2 mm by 6.6 mm (fig. S2).

After capturing and reverse-transcribing mRNA, we generated sequencing libraries based on amplification by in vitro transcription (fig. S3, A and B) (*7*, *8*). Comparison with data from RNA extracted and fragmented in solution revealed that ~95% of the genes found with one of the methods was also found with the other (fig. S3C). The correlation between the surface and in-solution libraries was r = 0.94, with even representation of genes having high or low expression (fig. S3D). Replicates of surface-based experiments of adjacent tissue sections showed a correlation of r = 0.97 (fig. S3E). Thus, cDNA synthesis from tissue with arrayed oligonucleotides on a surface is efficient and does not introduce bias compared with in-solution protocols (fig. S3F and table S1).

We sorted the RNA-seq data to its corresponding array features by using the spatial barcodes and aligned the tissue image with the features of the array, which enabled visualization and analyses. Examples of gene-expression patterns revealed by spatial transcriptomics and validation by in situ hybridization are shown in Fig. 2B and fig. S4, A to C. Transcripts expressed at very low levels, such as olfactory receptor mRNAs (*9*), were also detected with spatial transcriptomics (fig. S4D).

The number of genes (*10*) (Fig. 2C) and unique transcripts (fig. S5A) per individual feature varied between cell layers with different cell density (Fig. 2D and table S2). For the vast majority of genes, the coefficient of variation decreased as the average expression increased (fig. S5B). The number of

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